Week 6 Reference Sheet: Lists and Tuples

CSC-125-201 | Use this as a reference when completing Homework 6

Homework Topics

- Creating lists
- Indexing (positive and negative)
- Slicing
- List mutability
- List methods: append, remove, pop
- Built-in functions: len, sum, max, min
- Tuples and unpacking
- Lists vs tuples
- Writing pseudocode

Creating Lists

Ordered, mutable collections that hold any type of data.

Syntax	Example	Result
Empty list	`[]`	`[]`
With values	`["a", "b", "c"]`	`["a", "b", "c"]`
Numbers	`[1, 2, 3, 4, 5]`	`[1, 2, 3, 4, 5]`
Mixed types	`["text", 42, True]`	`["text", 42, True]`
From range	`list(range(5))`	`[0, 1, 2, 3, 4]`

Indexing

Access items by position. Starts at 0.

Expression	Given `fruits = ["apple", "banana", "cherry"]`	Result
`fruits[0]`	First item	`"apple"`
`fruits[2]`	Third item	`"cherry"`
`fruits[-1]`	Last item	`"cherry"`
`fruits[-2]`	Second to last	`"banana"`

Slicing

`list[start:end]` - end index NOT included.

Expression	Given `n = [10, 20, 30, 40, 50]`	Result
`n[1:4]`	Index 1 to 3	`[20, 30, 40]`
`n[:3]`	From start to index 2	`[10, 20, 30]`
`n[2:]`	From index 2 to end	`[30, 40, 50]`
`n[:]`	Full copy	`[10, 20, 30, 40, 50]`

Modifying Lists

Lists are mutable. Assign to an index to change an item.

```
fruits[1] = "mango"  # Changes "banana" to "mango"
```

List Methods

Method	What it does	Example	Result
`.append(x)`	Add x to end	`lst.append("d")`	`["a", "b", "c", "d"]`
`.remove(x)`	Remove first x	`lst.remove("b")`	`["a", "c", "d"]`
`.pop()`	Remove and return last	`lst.pop()`	Returns `"d"`, list is `["a", "c"]`
`.pop(i)`	Remove and return at index i	`lst.pop(0)`	Returns `"a"`, list is `["c"]`
`.insert(i, x)`	Insert x at index i	`lst.insert(0, "z")`	`["z", "c"]`
`.sort()`	Sort in place	`nums.sort()`	List becomes `[1, 2, 3]` (returns None)
`.reverse()`	Reverse in place	`nums.reverse()`	List becomes `[3, 2, 1]` (returns None)

Built-in Functions

Function	What it does	Example	Result
`len(lst)`	Number of items	`len([1,2,3])`	`3`
`sum(lst)`	Total of numbers	`sum([1,2,3])`	`6`
`max(lst)`	Largest value	`max([1,2,3])`	`3`
`min(lst)`	Smallest value	`min([1,2,3])`	`1`
`sorted(lst)`	New sorted list	`sorted([3,1,2])`	`[1, 2, 3]`
`x in lst`	Check membership	`2 in [1,2,3]`	`True`

Tuples

Immutable sequences created with parentheses. Cannot be changed after creation.

```
point = (10, 20)
x, y = point        # x = 10, y = 20
a, b = b, a         # Swap values
```

Lists vs Tuples

Feature	List `[]`	Tuple `()`
Mutable	Yes	No
Use when	Data changes	Data is fixed
Examples	Shopping cart, scores	Coordinates, colors
Methods	Many (append, remove, sort...)	Only count, index

Pseudocode Basics

Structured English describing program logic. Plan before coding.

Verb	Meaning	Python equivalent

GET	Receive input	`input()`
SET/CREATE	Store a value	`x = value`
CALCULATE	Do math	`result = a + b`
DISPLAY	Show output	`print()`
IF/ELSE	Make decision	`if/else`

Common Errors

Error	Cause	Fix
`IndexError`	Index out of range	Check with `len()` first
`ValueError`	`.remove()` item not in list	Check with `in` first
`TypeError`	Modifying a tuple	Use a list if data needs to change
Logic error	Forgetting end is excluded in slicing	`[1:4]` gets indices 1, 2, 3
Logic error	Assigning `.sort()` result	`.sort()` returns None, use `sorted()` for new list

Resources

- Corey Schafer - Python Lists Tutorial
- W3Schools - Python Lists
- Real Python - Lists and Tuples